

3.3.8 Aldehydes and ketones (A-level only)

Aldehydes, ketones, carboxylic acids and their derivatives all contain the carbonyl group which is attacked by nucleophiles. This section includes the addition reactions of aldehydes and ketones.

Content	Opportunities for skills development
Aldehydes are readily oxidised to carboxylic acids. Chemical tests to distinguish between aldehydes and ketones including Fehling's solution and Tollens' reagent. Aldehydes can be reduced to primary alcohols, and ketones to secondary alcohols, using NaBH_4 in aqueous solution. These reduction reactions are examples of nucleophilic addition. The nucleophilic addition reactions of carbonyl compounds with KCN, followed by dilute acid, to produce hydroxynitriles. Aldehydes and unsymmetrical ketones form mixtures of enantiomers when they react with KCN followed by dilute acid. The hazards of using KCN. Students should be able to: • write overall equations for reduction reactions using $[\text{H}]$ as the reductant • outline the nucleophilic addition mechanism for reduction reactions with NaBH_4 (the nucleophile should be shown as H^-) • write overall equations for the formation of hydroxynitriles using HCN • outline the nucleophilic addition mechanism for the reaction with KCN followed by dilute acid • explain why nucleophilic addition reactions of KCN, followed by dilute acid, can produce a mixture of enantiomers.	AT b, d and k PS 2.2 Students could carry out test-tube reactions of Tollens' reagent and Fehling's solution to distinguish aldehydes and ketones.